

5Q) Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code)

```
sai@kornypc: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
int main(){  
    int pipefd[2];  
    pipe(pipefd);  
    if (fork() == 0){  
        close(pipefd[0]);  
        dup2(pipefd[1], STDOUT_FILENO);  
        close(pipefd[1]);  
        execlp("ls", "ls", "-l", NULL);  
        perror("exec ls failed");  
    }else{  
        if (fork() == 0) {  
            close(pipefd[1]);  
            dup2(pipefd[0], STDIN_FILENO);  
            close(pipefd[0]);  
            execlp("grep", "grep", ".c", NULL);  
            perror("exec grep failed");  
        }  
        else  
        {  
            close(pipefd[0]);  
            close(pipefd[1]);  
  
            wait(NULL);  
            wait(NULL);  
        }  
    }  
  
    return 0;  
}
```

Output)

```
sai@kornypc:~$ ./p5
drwxr-xr-x 3 sai sai 4096 Aug  1 08:09 MyProject
-rw-r--r-- 1 sai sai  391 Aug 12 07:02 bye.c
drwxr-xr-x 2 sai sai 4096 Jul 27 05:48 cd
-rwxr-xr-x 1 sai sai 16048 Aug 29 08:18 fifo_create
-rw-r--r-- 1 sai sai  218 Aug 29 08:20 fifo_create.c
-rw-r--r-- 1 sai sai  434 Aug 29 08:33 fifo_read.c
-rw-r--r-- 1 sai sai  292 Aug 29 08:27 fifo_write.c
-rw-r--r-- 1 sai sai  797 Aug 14 07:03 fork.c
-rw-r--r-- 1 sai sai  738 Aug 14 04:35 hello.c
-rw-r--r-- 1 sai sai  715 Aug 14 04:36 hello.c'
-rw-r--r-- 1 sai sai  127 Aug  1 04:14 helo.c
-rw-r--r-- 1 sai sai  618 Aug 29 09:57 ls_grep.c
-rw-r--r-- 1 sai sai 1653 Aug 12 06:37 main.c
-rw-r--r-- 1 sai sai 1029 Aug 14 07:44 p2.c
-rw-r--r-- 1 sai sai 1092 Aug 14 07:36 p4.c
-rw-r--r-- 1 sai sai  871 Aug 31 05:35 p5.c
-rw-r--r-- 1 sai sai  722 Aug 29 08:15 pipe.c
-rw-r--r-- 1 sai sai  475 Aug 12 06:43 pr.c
-rw-r--r-- 1 sai sai  672 Aug 14 05:30 pr1.c
-rw-r--r-- 1 sai sai  717 Aug 14 04:46 pr1.co
-rw-r--r-- 1 sai sai  712 Aug 14 04:47 pr1c
-rw-r--r-- 1 sai sai  659 Aug 14 06:54 pr2.c
-rw-r--r-- 1 sai sai  258 Aug 17 04:49 program.c
-rw-r--r-- 1 sai sai  391 Aug 12 07:00 program.c.save
-rw-r--r-- 1 sai sai  150 Aug  3 04:42 program1.c
-rw-r--r-- 1 sai sai 1092 Aug  7 05:40 program2.c
-rw-r--r-- 1 sai sai  299 Aug 29 09:28 signal1.c
-rw-r--r-- 1 sai sai  168 Aug 31 05:10 signal2.c
sai@kornypc:~$ |
```